

# ARAVINTH K

## Senior Embedded Firmware Engineer

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### PROFILE SUMMARY

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Senior Embedded Firmware Engineer with 5+ years of experience in embedded firmware development, debugging, integration and validation across industrial automation and consumer electronics domains. Skilled in Embedded C/C++, Zephyr RTOS, Embedded Linux and communication protocols including CAN, SPI, I2C, UART and MQTT. Delivered firmware for three commercial Corsair sim-racing products and received several Bosch awards for quality and delivery excellence.

### TECHNICAL SKILLS

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**Programming Languages:** Embedded C, C++, Shell Scripting

**RTOS & Operating Systems:** Zephyr RTOS, Embedded Linux

**Microcontrollers:** STM32, TI TIVA C, NXP LPC Series, Atmel AVR/SAM, Microchip PIC

**Communication Protocols:** CAN, UART, SPI, I2C, RS232, RS485, MQTT

**Wireless & Connectivity:** Bluetooth, Wi-Fi, GPS, GSM, RFID

**Embedded Linux:** System Programming, POSIX Threads, Inter-Process Communication (IPC), Character Device Drivers

**Embedded Systems:** Firmware Development, Firmware Validation, Embedded Software Design, System Integration, Debugging, Root Cause Analysis

**Testing & Quality:** Unit Testing (Ceedling), Integration Testing, System Testing, Static Code Analysis (Cppcheck)

**Tools & IDEs:** Git, Gerrit, Version control, JIRA, GDB, JTAG, Oscilloscope, Logic Analyzer, VS Code, Keil MDK, MPLAB X, Atmel Studio, Code Composer Studio (CCS)

### WORK EXPERIENCE

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**Analyst (Senior Embedded Firmware Engineer) | Bosch Global Software Technologies Pvt. Ltd.** Feb 2023 – Present  
Coimbatore, India

- Developed firmware for three commercial Corsair sim-racing products using Zephyr RTOS, including Handbrake, PVGT Porsche and Formula V3 Steering Wheel.
- Resolved 21+ defects across 4 Corsair product lines through defect resolution, root cause analysis, regression testing, as well as feature enhancements, improving product stability and reliability.
- Led and mentored a team of 2 junior firmware engineers, providing technical guidance, task planning, code reviews, and project onboarding support.

**Embedded Software Developer | Kalycito**  
Coimbatore, India

Apr 2022 – Jan 2023

- Supported C++ application software on Linux for EdgeXConnect, a configurable no-code OPC UA Server platform for industrial IoT environments.
- Performed defect fixes, OPC UA node modifications, and release validation activities in an Agile development environment.
- Improved code organization and maintainability by restructuring the project using Git submodules and contributing to release validation activities.

- Sole firmware engineer for a Three-Layer Conveyor System (textile automation) delivered firmware development, system integration, testing, and on-site deployment end-to-end in under 4 months.
- Calibrated and fine-tuned stepper motor control firmware for an Automatic Cone Winding Machine, implementing bug fixes and enhancements to improve automation performance and system reliability.
- Designed and validated a pulse oximeter proof-of-concept using the MAX30100 sensor, implementing SpO<sub>2</sub> and heart-rate measurement algorithms with  $\pm 2\%$  accuracy.

## KEY PROJECTS

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### Corsair Sustainability, PVGT Porsche & Formula V3 Steering Wheels

Bosch · 2024 – Present

Stack: *Embedded C, Zephyr RTOS, LPC, CAN, SPI, I2C*

- Delivered sustainability support for multiple Corsair (Endor) sim-racing products through defect resolution, root cause analysis, regression testing and firmware enhancements.
- Developed and customized firmware for the PVGT Porsche and Formula V3 Steering Wheels using Zephyr RTOS by extending an existing firmware framework. Implemented product-specific features, performed product integration and system-level testing, and supported release activities.

### Corsair Handbrake — Sim Racing Peripheral Firmware

Bosch · 2023 - 2024

Stack: *Embedded C, Zephyr RTOS, NXP LPC, CAN Bus, Sensor Interface, EEPROM*

- Developed Zephyr RTOS firmware for a commercial sim-racing handbrake, implementing HX717 sensor acquisition, CAN communication, EEPROM-based profile storage and bootloader integration.
- Coordinated product release activities, including wheelbase compatibility validation, regression testing, and launch sign-off.

### EdgeXConnect — Industrial No-Code OPC UA Server

Kalycito · 2022 – 2023

Stack: *C++, Linux, OPC UA SDK, CMake*

- Maintained and enhanced a C++ OPC UA server application on Linux, resolving defects and contributing to integration and system testing.

## AWARDS & RECOGNITION

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### Best Tyro — Bosch

*Recognized for quick adaptation, strong commitment, and high productivity within a short ramp-up period.*

### Catalyst — Bosch

*Excellent defect identification in Corsair Steering Wheel firmware ahead of the product launch schedule.*

### Growth Driver — Bosch

*Outstanding contribution to the Corsair firmware sustainability initiative.*

## EDUCATION

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### B.E — Electronics and Communication Engineering

2013 – 2017

Nandha College of Technology, Erode